CS228: Introduction to Deep Learning

Lecture 5: activation and error functions

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The perceptron algorithm

- Initialize W (random or simply 0)
- For each training example, **update W only if it's misclassified**
  - Positive sample misclassified:

    $$\mathbf{W} \leftarrow \mathbf{W} + X_i$$

  - Negative sample misclassified:

    $$\mathbf{W} \leftarrow \mathbf{W} - X_i$$

Generally:

$$\mathbf{W} \leftarrow \mathbf{W} + y_i X_i$$

$$(y_i \in \{-1, 1\})$$

Whiteboard: demo

Convergence

As long as the classes are **linearly separable**:

The perceptron algorithm is **guaranteed to converge to perfect classification** within a number of misclassifications

Convergence

Bounded scale:

$$\|X_i\| \leq R$$

$$\|W^*\| = 1$$

- Linearly separable:

$$\forall 1 \leq i \leq n, \;\; y_i(\mathbf{W}^* \cdot X_i) > \gamma \; (\gamma > 0)$$

- Maximum number of misclassifications: $\left(\frac{R}{\gamma}\right)^2$

- If not linearly separable, the algorithm does not converge
- Cannot be applied to learn MLPs

Whiteboard: proof (optional)

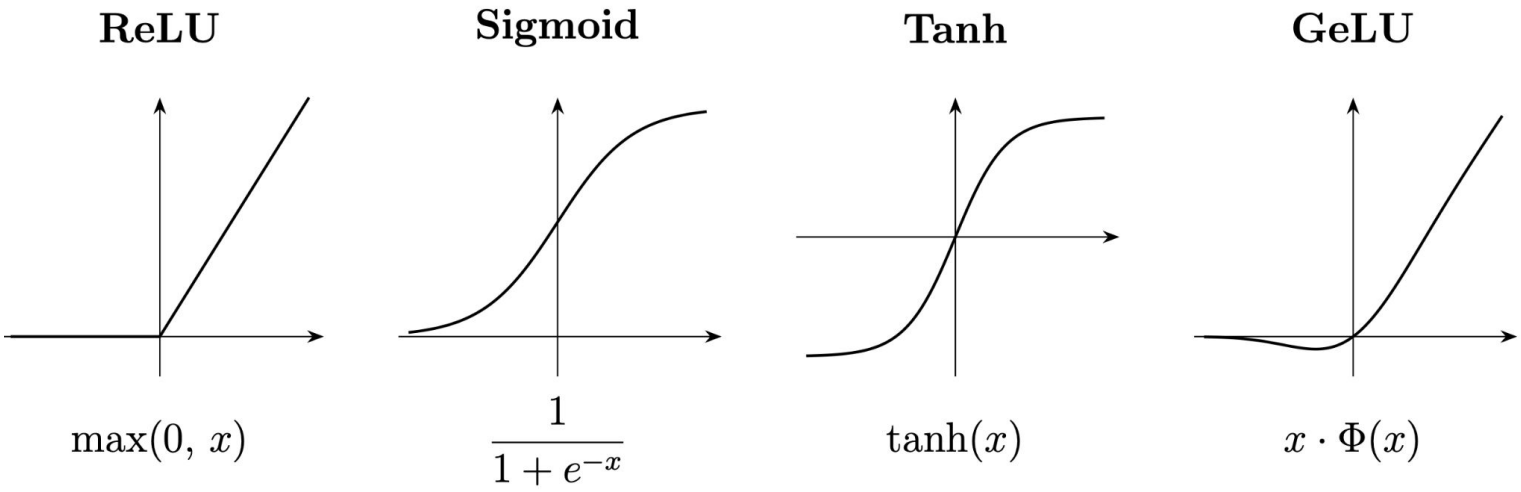
Issues with the original perceptron

- Practical problems are typically not linearly separable and require deep models
- The activation function (threshold) is not continuous
- The 0-1 error function is not continuous

- Use continuous and differentiable functions
  - To know if we are going the right way with some small change

Whiteboard: non-continous activation and error functions

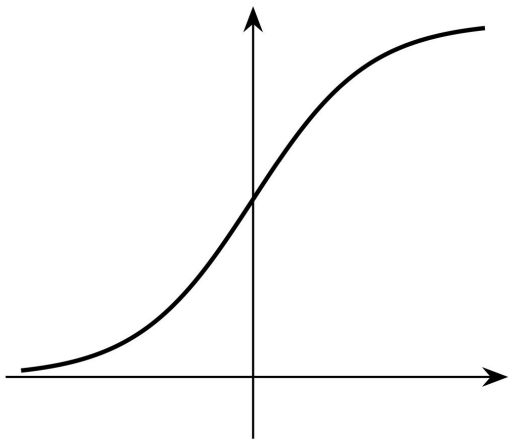
Smooth activation functions



- Differentiable almost everywhere
  - “Subderivatives” at non-differentiable corners (interval [a, b] from one-sided limits)

Sigmoid for probabilities

Sigmoid



$$\frac{1}{1+e^{-x}}$$

Sigmoid can model P(y=1|x)

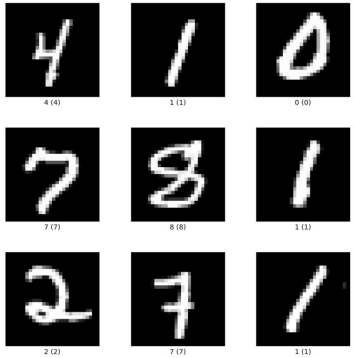
Whiteboard: sigmoid for P(y=1|x) in 1D

Softmax

A special activation that is applied to multiple neurons

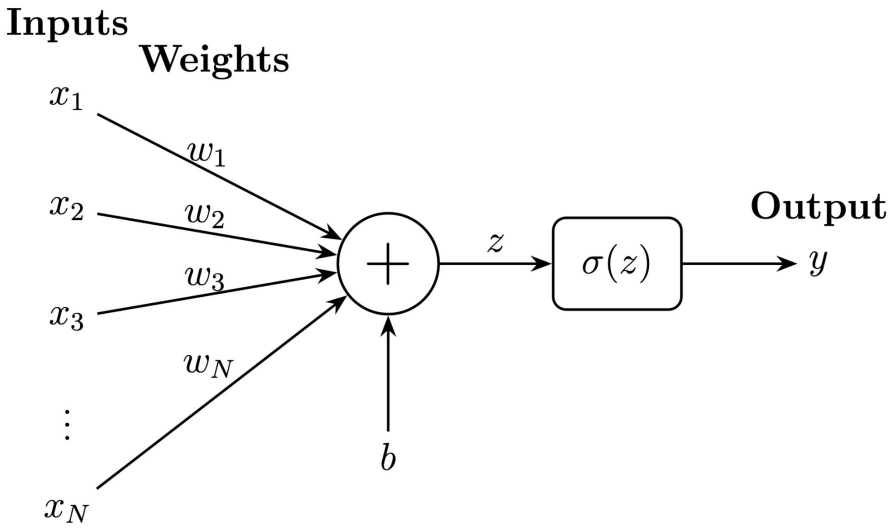
$$y_i = \mathrm{Softmax}(z) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \qquad \sum_i y_i = 1$$

Model probability P(y=i | x) in multi-class classification settings



Equivalent to sigmoid in binary classification

Differentiable model



$$\frac{dy}{dz} = \sigma'(z)$$

$$\frac{dy}{dw_i} = \frac{dy}{dz}\frac{dz}{dw_i} = \sigma'(z)x_i$$

$$\frac{dy}{dx_i} = \frac{dy}{dz}\frac{dz}{dx_i} = \sigma'(z)w_i$$

$$\frac{dy}{db} = \frac{dy}{dz}\frac{dz}{db} = \sigma'(z)$$

- y is differentiable with both inputs and parameters
- The derivatives tell us how the output changes for small changes in input or parameters

Differentiable error function

$$\text{empirical_error} = \sum_{i=1}^{n} \text{error}(y_i, g_i)$$

On i-th example: Output Ground-truth

- The error function should be differentiable
- Typically called “*loss*”

Loss functions

L2 divergence (squared Euclidean distance) for output vectors:

$$\mathrm{Loss}(y_i, g_i) = \frac{1}{2}\|y_i - g_i\|^2$$

Kullback Leibler (KL) divergence:

$$\mathrm{KL}(y_i, g_i) = \sum_j g_{ij} \log \frac{g_{ij}}{y_{ij}} = \sum_j g_{ij} \log g_{ij} - \sum_j g_{ij} \log y_{ij}$$

- Both reaches the minimum when y=g
- KL rises much faster when y is away from g

Whiteboard: derivatives and plot (binary)

Loss functions

- Multi-class classification:
  - One hot vector for g: [0 0 ... 1 ... 0 0 0]
  - g_c = 1 means the ground-truth class is c

- KL divergence: $\mathrm{Loss}(y_i, g_i) = -\log y_{ic_i}$

KL v.s. cross-entropy

KL divergence: $\mathrm{KL}(y_i, g_i) = \sum_j g_{ij} \log \frac{g_{ij}}{y_{ij}} = \sum_j g_{ij} \log g_{ij} - \sum_j g_{ij} \log y_{ij}$

Difference when we use y (prediction) to approximate g (ground-truth)

Cross entropy: $H(y_i, g_i) = -\sum_j g_{ij} \log y_{ij} \quad H(y_i, g_i) = \mathrm{KL}(y_i, g_i) + H(g_i, g_i)$

Total number of bits in expectation (add entropy of g)

KL and cross entropy are equivalent when g is one-hot

Usually a cross-entropy loss is used for classification problems